

PulseChat Project Report

JWT WebSocket Chat App is a full-stack real-time chat application built with Spring Boot, React, PostgreSQL, JWT authentication, and STOMP/WebSocket messaging.

Spring Boot

React + Vite

PostgreSQL

JWT

WebSocket

Project Overview

The project provides secure account registration, email OTP verification, login, protected frontend routes, profile management, private chat, group chat, media sharing, message replies, group polls, and real-time presence updates.

The PulseChat landing page uses a restrained natural hero image and minimal copy so the first screen feels clean and professional.

Frontend

React 19, Vite, React Router DOM, Redux Toolkit, Axios, and custom CSS.

Backend

Spring Boot, Spring Security, Spring WebSocket, Spring Data JPA, and layered services.

Database

PostgreSQL persistence for users, messages, groups, uploads, polls, and notifications.

Security

JWT access/refresh token flow, HttpOnly cookies, BCrypt password hashing, and protected APIs.

Key Features

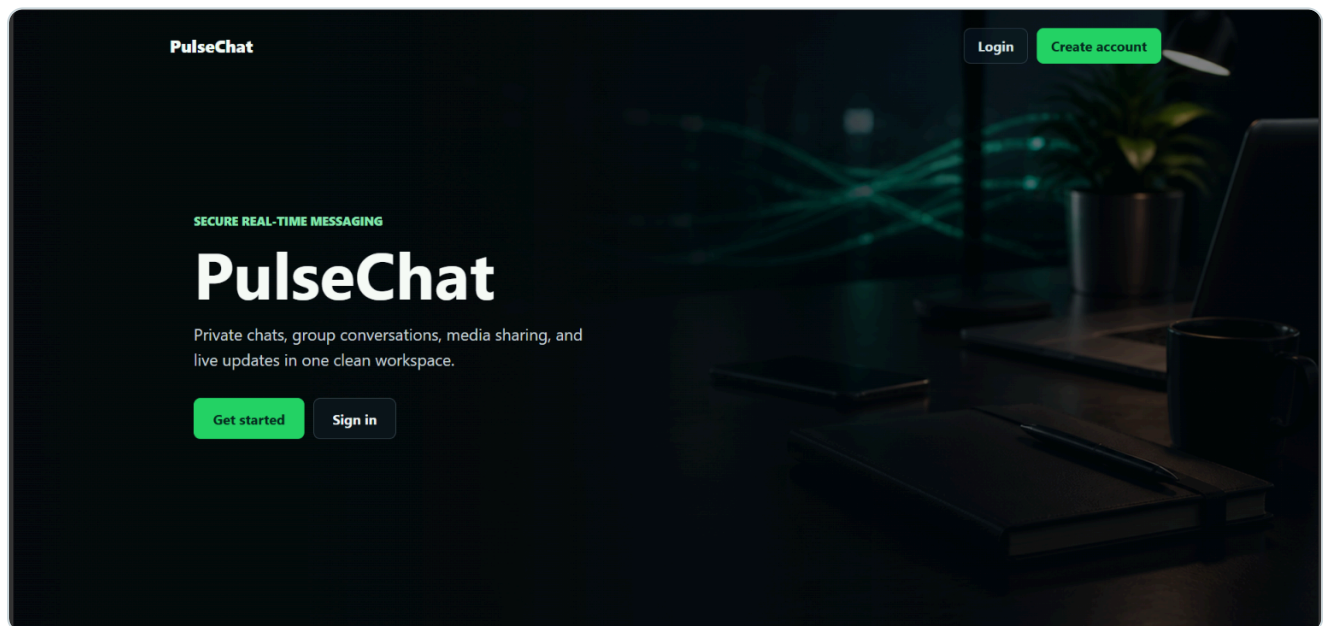
- User registration with email OTP verification before login.
- JWT-based authentication with refresh handling and protected frontend routes.
- Private one-to-one chat with persisted messages and real-time delivery.
- Group chat with members, roles, mentions, replies, seen information, and polls.
- Profile, cover, group, image, and video upload support with validation.

System Architecture

Layer	Responsibility
React Frontend	Routes, authentication state, chat UI, profile screens, API calls, and WebSocket subscriptions.
Spring Boot API	REST controllers, auth endpoints, upload handling, business rules, and WebSocket publish/subscribe flow.
Service Layer	Validation, authorization checks, message/group workflows, OTP handling, and profile logic.
PostgreSQL	Persistent data store for users, chats, messages, groups, polls, uploads, and notification data.

Landing Page UI

The screenshot below is the current landing page used in the README.



Local Setup Summary

- Run the Spring Boot backend from the **backend** folder with Maven.
- Run the React frontend from the **frontend** folder with **npm run dev**.
- Configure database, mail, and JWT values through environment-specific backend settings.